



DW01_sub

Subtractor

Version, STAR, and myDesignWare Subscriptions: [IP Directory](#)

Features and Benefits

- Parameterized word length
- Carry-in and carry-out signals

Revision History

Description

DW01_sub is a two-input subtractor. DW01_sub subtracts two operands A and B with a carry-in (CI) to produce the output DIFF with a carry-out (CO).

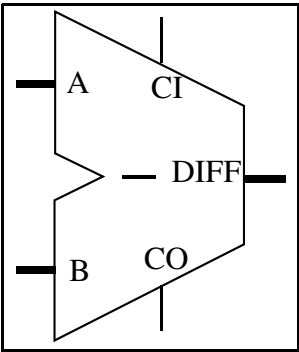


Table 1-1 Pin Description

Pin Name	Width	Direction	Function
A	<i>width</i> bits	Input	Input data
B	<i>width</i> bits	Input	Input data
CI	1 bit	Input	Carry-in
DIFF	<i>width</i> bits	Output	Difference of $A - B - CI$
CO	1 bit	Output	Carry-out

Table 1-2 Parameter Description

Parameter	Values	Description
width	≥ 1	Word length of A, B, and DIFF

Table 1-3 Synthesis Implementations^a

Implementation Name	Function	License Feature Required
rpl	Ripple-carry synthesis model	none
cla	Carry-look-ahead synthesis model	none
pparch	Delay-optimized flexible parallel-prefix	DesignWare
apparch	Area-optimized flexible architecture that can be optimized for area, for speed, or for area, speed	DesignWare

- a. During synthesis, Synopsys synthesis tools select the appropriate architecture for your constraints. Alternatively, you may use the `set_implementation` command to choose one specific implementation from those listed in this table. For details, see the documentation for your Synopsys synthesis tool.

Table 1-4 Simulation Models

Model	Function
DW01.DW01_SUB_CFG_SIM	Design unit name for VHDL simulation
dw/dw01/src/DW01_sub_sim.vhd	VHDL simulation model source code
dw/sim_ver/DW01_sub.v	Verilog simulation model source code

Related Topics

- [Math – Arithmetic Overview](#)
- [DesignWare Building Block IP User Guide](#)

HDL Usage Through Operator Inferencing - VHDL

```
library IEEE;
use IEEE.std_logic_1164.all;
use IEEE.std_logic_arith.all;

entity DW01_sub_oper is
    generic(wordlength: integer := 8);
    port(in1,in2 : in STD_LOGIC_VECTOR(wordlength-1 downto 0);
         diff : out STD_LOGIC_VECTOR(wordlength-1 downto 0));
end DW01_sub_oper;

architecture oper of DW01_sub_oper is
    signal in1_signed, in2_signed, diff_signed: SIGNED(wordlength-1 downto 0);
begin
    in1_signed <= SIGNED(in1);
    in2_signed <= SIGNED(in2);
    -- infer the "-" subtraction operator
    diff_signed <= in1_signed - in2_signed;
    diff <= STD_LOGIC_VECTOR(diff_signed);
end oper;
```

HDL Usage Through Operator Inferencing - Verilog

```
module DW01_sub_oper(in1,in2,diff);  
    parameter wordlength = 8;  
  
    input [wordlength-1:0] in1,in2;  
    output [wordlength-1:0] diff;  
  
    assign diff = in1 - in2;  
endmodule
```

HDL Usage Through Component Instantiation - VHDL

```
library IEEE,DWARE;
use IEEE.std_logic_1164.all;
use DWARE.DWpackages.all;
use DWARE.DW_foundation_comp.all;

entity DW01_sub_inst is
  generic ( inst_width : NATURAL := 8 );
  port ( inst_A      : in std_logic_vector(inst_width-1 downto 0);
        inst_B      : in std_logic_vector(inst_width-1 downto 0);
        inst_CI     : in std_logic;
        DIFF_inst   : out std_logic_vector(inst_width-1 downto 0);
        CO_inst     : out std_logic );
end DW01_sub_inst;

architecture inst of DW01_sub_inst is
begin
  -- Instance of DW01_sub
  U1 : DW01_sub
    generic map ( width => inst_width )
    port map ( A => inst_A,  B => inst_B,  CI => inst_CI,
              DIFF => DIFF_inst,  CO => CO_inst );
end inst;

-- pragma translate_off
configuration DW01_sub_inst_cfg_inst of DW01_sub_inst is
  for inst
    end for; -- inst
end DW01_sub_inst_cfg_inst;
-- pragma translate_on
```

HDL Usage Through Component Instantiation - Verilog

```
module DW01_sub_inst( inst_A, inst_B, inst_CI, DIFF_inst, CO_inst );

    parameter width = 8;

    input [width-1 : 0] inst_A;
    input [width-1 : 0] inst_B;
    input inst_CI;
    output [width-1 : 0] DIFF_inst;
    output CO_inst;

    // Instance of DW01_sub
    DW01_sub #(width)
        U1 ( .A(inst_A), .B(inst_B), .CI(inst_CI),
            .DIFF(DIFF_inst), .CO(CO_inst) );

endmodule
```

Revision History

For notes about this release, see the [DesignWare Building Block IP Release Notes](#).

For lists of both known and fixed issues for this component, refer to the [STAR report](#).

For a version of this datasheet with visible change bars, click [here](#).

Date	Release	Updates
September 2024	DWBB_202409.0	■ Updated date and version; no technical changes
December 2023	DWBB_202312.0	■ Updated date and version; no technical changes
March 2019	DWBB_201903.0	■ Removed Table 1-4, “Obsolete Synthesis Implementations”
January 2019	DWBB_201806.5	■ Updated example in “ HDL Usage Through Component Instantiation - VHDL ” on page 5 ■ Added this Revision History table and the document links on this page

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